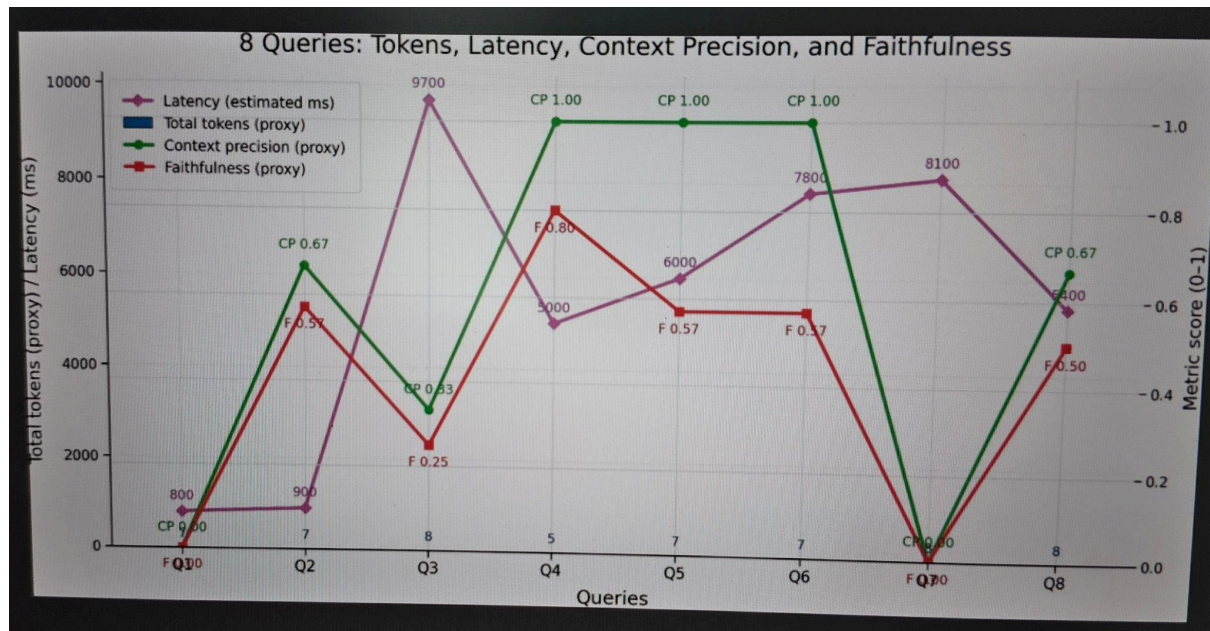


Latency & Retrieval Analysis Summary (Q1–Q8):



1. Q3 is a reasoning bottleneck, not a token issue

Despite having only ~8 tokens, Q3 shows the highest latency (~9700 ms). This indicates that reasoning complexity, not output length, is the dominant driver of delay.

2. Strong retrieval does not guarantee low latency

Q4, Q5, and Q6 achieve perfect context precision (1.00), yet latency increases from ~5000 ms to ~7800 ms. This shows that even with accurate retrieval, generation time still impacts performance.

3. Simple queries are significantly faster

Q1 and Q2 (7 tokens each) complete in under ~1000 ms latency. These queries require minimal reasoning and have low generation cost.

4. Q7 is the most difficult case

Q7 shows high latency (~8100 ms) with near-zero context precision and weak faithfulness. This indicates a combination of semantic mismatch and higher reasoning complexity.

5. Faithfulness depends on retrieval, not speed

Faithfulness generally aligns with context precision and answer relevance, but does not correlate with latency. This confirms that generation effort and retrieval quality are independent factors.